Python Project

Title:

Hangman (Word Guess Game)

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Subject:

Python Programming

Aim

To design and develop a beginner-friendly Hangman game in Python that demonstrates fundamental programming concepts such as randomization, decision making, looping, arrays (lists), strings, and functions.

Procedure

Step 1: Random Word Selection

- A list of words is defined.
- The random.choice() function selects one word at random for the player to guess.

Step 2: Game State Initialization

- The word is represented as underscores (_) for each letter.
- A set is used to track already guessed letters.
- The player is given a fixed number of attempts (6).

Step 3: Looping & Decision Making

- A while loop runs until the player either guesses the word or runs out of attempts.
- Decision making (if/else) checks whether the guessed letter is correct, already used, or wrong.
- Correct guesses update the word display; wrong guesses reduce attempts.

Step 4: Strings & Arrays

- Strings are used for player input and word comparison.

- Arrays (lists) store the current guessed state of the word.
- The game prints the updated word after each guess.

Step 5: Functions

- The game logic is encapsulated in the hangman() function.
- The __main__ block ensures the game runs when executed directly.

Code

```
import random

def hangman():
    words = ["python", "data", "science", "hangman", "project"]
    word = random.choice(words)

    guessed = ["_"] * len(word)
    attempts = 6
    used_letters = set()

    print("Hangman Game")

    while attempts > 0 and "_" in guessed:
        print("Word:", " ".join(guessed))
        print("Attempts left:", attempts)

        guess = input("Guess a letter: ").lower()

        if guess in used_letters:
            print("You already tried that letter.")
            continue

        used_letters.add(guess)

        if guess in word:
            for i, letter in enumerate(word):
                if letter == guess:
                    guessed[i] = guess
            print("Correct!")
        else:
            attempts -= 1
            print("Wrong guess!")

    if "_" not in guessed:
        print("You win! The word was:", word)
    else:
        print("Game over! The word was:", word)
```

```
if __name__ == "__main__":
    hangman()
```

Output

```
Hangman Game
Word: _ _ _ _ _
Attempts left: 6
Guess a letter: a
Correct!
Word: _ a _ a _
Attempts left: 6
Guess a letter: z
Wrong guess!
Word: _ a _ a _
Attempts left: 5
```

Result

The Hangman game was successfully developed in Python.
It demonstrates:

- Randomization for word selection
- Decision making for correct/incorrect guesses
- Looping for repeated gameplay
- Arrays (lists) for storing guessed letters
- Strings for input and word handling
- Functions for modular design